

JASH MEHTA

✉ jashmehta3300 🌐 jashmehta3300.github.io 📄 [jashmehta3300](#)

EDUCATION

Dwarkadas J. Sanghvi College of Engineering 2018 - 2022(*expected*)
(University of Mumbai) Overall GPA: 9.60/10
 Bachelor of Engineering (B.E.) in Computer Engineering
 Machine Learning, Artificial Intelligence, Natural Language Processing, Applied Mathematics, Algorithms & Data Structures, Database Management

EXPERIENCE

IIT Gandhinagar (LINGO) May 2021 - Present
Research Intern Advisor: [Dr. Mayank Singh](#)

- Demystifying automated peer-review generators and evaluating their robustness to input perturbations.
- We formulate desiderata for an ideal review generator system, provide a public leaderboard and framework for unified & comprehensive measurement of their performance.
- Participated in presentations and discussions at LINGO (Computational Linguistics Group)

Unicode Research Aug 2020 - Present
Research Student Advisor: [Jennifer Hill](#), [Swapneel Mehta](#)

- **Project:** Estimating the causal impact of non-expert mentorship on career outcomes of Indian students.
- **Teaching Assistant:** Machine Learning Course [UMLSC 2021](#), supported by **Google Research India**.
- Member of [Shalizi-Stats](#) reading group which deals with *Bayesian Statistics* and book by *Cosma Shalizi*.

Dwarkadas J. Sanghvi College of Engineering Jan 2021 - June 2021
Research Assistant Advisor: [Dr. Ramchandra Mangrulkar](#)

- Made a project dealing with application of Federated Learning for highly sensitive medical data.
- Worked on a research project to identify Spear Phishing using low computational NLP approaches.
- Published 2 chapters for CRC Press in the domains of Federated Learning & NLP

Feople Org July 2020 - Dec 2020
Machine Learning Intern

- Devised dynamic pricing strategy for a restaurant which helped in improving the sales by 28%.
- Worked with ML team for development of recommender system pipeline and other internal ML systems.

Margosatree Technologies Jan 2020 - June 2020
Fullstack Web Developer margosatree.com

- Created a tool to gain insights from dynamic data coming from the manufacturing machines.
- Worked on a diverse array of client and internal projects like inventory management system and application to track employee productivity.

PROJECTS

A Federated Approach to Predict Emojis in Hindi Tweets Advisor: [Dr. Zeerak Talat](#)

- Cost sensitive learning and SMOTE for imbalanced emoji data using FedProx for training.
- Curating a dataset of 1M tweets to predict emojis resource-constrained language.

IndicFed: A Federated Approach for Indic Languages

Advisor: [Prof. Sudhir Bagul](#)

- Training sentiment analysis models for 4 resource-poor languages using FL to address data scarcity issue.
- Outperforming server-trained counterparts and transformer models having 10x more parameters.

Generalisable Hate Speech Detection using Federated Learning

Advisor: [Dr. Zeerak Talat](#)

- Studying cross-dataset generalization using FL to capture different complexities of hate speech effectively
- The experiments consider the models AWD-LSTM, FNet, DistilBERT, DistilRoberta, etc.

FedHealth

Advisor: [Prof. Lynette D'Mello](#)

- Final year project to train models on EHR data using multi-modular blockchain-based FL architecture.
- Creating representations for personalized prescription based on user reviews using Med-BERT embeddings.

PUBLICATIONS

- [1] Deep Gandhi*, **Jash Mehta***, and Pranit Bari, “Ablation analysis of seq2seq models and vanilla transformers for spanish to english translation,” in *Proceedings of the 3rd International Conference on Advances in Distributed Computing and Machine Learning (ICADCML'22)*, Springer Nature, Accepted.
- [2] Deep Gandhi, **Jash Mehta**, Nemil Shah, and Dr.Ramchandra Mangrulkar, “Federated Learning for Brain Tumor Segmentation on the Cloud,” in *Cloud Computing Technologies for Smart Agriculture and Healthcare*, pp. 261–278, Chapman and Hall/CRC, 2021.
- [3] **Jash Mehta**, Deep Gandhi, Govind Thakur, and Pratik Kanani, “Music genre classification using transfer learning on log-based mel spectrogram,” in *2021 5th International Conference on Computing Methodologies and Communication (ICCMC)*, pp. 1101–1107, IEEE, 2021.
- [4] Deep Gandhi, **Jash Mehta**, and Dr.Ramchandra Mangrulkar, “Detection of Spear Phishing using Natural Language Processing,” in *Cyber Security Threats and Challenges facing Human Life*, ch. 9, Chapman and Hall/CRC, *Accepted*.

TECHNICAL STRENGTHS

Languages	Python, R, Javascript, C
Libraries/Frameworks	PyTorch, Huggingface, fast.ai, Flower, Flask, Node.js, React.js, GraphQL
Databases	SQL, MongoDB, Redis, PostgreSQL
Tools	Git, Jupyter, Docker, Bash, AWS, Azure, L ^A T _E X

CO-CURRICULAR ACTIVITIES & ACHIEVEMENTS

- Awarded **INSPIRE scholarship**, Top 1% candidates in Higher Secondary Certificate, 2018
- Coding member of DJS Arya, a cansat making team which achieved **24th rank** (Worldwide) in PDR and **34th rank** (Worldwide) in 2018-19
- Built predictive model for automotive componentpart failure for a Big 4 consultancy firm under the guidance of *Prof. Kriti Shivastava*
- Member of [Shalizi-Stats](#) reading group which deals with [Bayesian Statistics](#) and focuses on the book [Advanced Data Analysis from an Elementary Point of View](#) by Cosma Shalizi.